Wavelet Processing

- Fourier orthonormal basis
- Compact support orthonormal basis: Haar functions
- Wavelet theory.
- Fast Wavelet transform

Basis
Synthesis Formula

- The set of K vectors $W = \{\mathbf{x}^{(K)}\}_{k=1\cdots K}$ from a subspace P is a base of this subspace if :
  - The set W is linearly independent
  - The span covers completely P: span{W}=P and we can write for any $\mathbf{y} \in P$ could be written as a linear combination of $\{\mathbf{x}^{(K)}\}_{k=1\cdots K}$

$$\mathbf{y} = \sum_{k=1}^{K} \alpha_k \mathbf{x}^{(K)}$$

- Orthogonal and orthonormal basis: in this case the set $W = \{\mathbf{x}^{(K)}\}_{k=1\cdots K}$ we have the constraint:

  $$\langle \mathbf{x}^{(i)}, \mathbf{x}^{(j)} \rangle = \delta[i-j]$$

  when i=j the scalar product is equal1 otherwise is equal 0.
- A set of N orthogonal vectors in a N dimension space is a base of this space.

Properties of orthonormal basis

- $W = \{\mathbf{x}^{(K)}\}_{k=1\cdots K}$ is an orthonormal basis of a subspace P, the properties in this case are:
- **Analysis Formula**: The coefficients α_k are obtained for any vector $\mathbf{y}$ of the subspace P :

$$\alpha_k = \langle \mathbf{x}^{(k)}, \mathbf{y} \rangle$$

- **Parseval Identity:** $\|\mathbf{y}\|^2 = \sum_{k=1}^{K} |\langle \mathbf{x}^{(k)}, \mathbf{y} \rangle|^2$

exprime the energy conservation

Properties of orthonormal basis

- **Best approximation:** P (K dimensions) is a subspace of V (N dimensions), we try to approximate a vector $\mathbf{y} \in V$ with a linear comb. of vector of the subspace P.
  We need to define the best linear approximation $\hat{\mathbf{y}} \in P$ of a vector $\mathbf{y} \in V$ that minimise the norm $||\mathbf{y} - \hat{\mathbf{y}}||$. This best approximation is obtained projecting **y** onto the orthonormal basis of P as:

$$\hat{\mathbf{y}} = \sum_{k=1}^{K} \alpha_k \cdot \mathbf{x}^{(k)}$$

  - The error $\mathbf{d} = \mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to the approximation $\hat{\mathbf{y}}$
  - The approximation minimize the square norm of **d**

Finite Euclidian Spaces

- For exemple the Fourrier basis $\{\mathbf{w}^{(k)}\}$:
  $\mathbf{w}_n^{(k)} = e^{j\frac{2\pi}{N}\mathbf{n}k}$ is a complex orthogonal basis.
- Since this basis functions have $\|w_n^{(k)}\|_2 = \sqrt{N}$ to conserve the energy after transform we need to scale the basis as $\mathbf{w}_n^{(k)} = \frac{1}{\sqrt{N}} e^{j\frac{2\pi}{N}\mathbf{n}k}$
- if we arrange a transformation matrix M (NxN) containing at each line k the n line vector $(\mathbf{w}_n^{(k)})^*$ we could compute the vector α with analysis formula:

$$\alpha = My$$

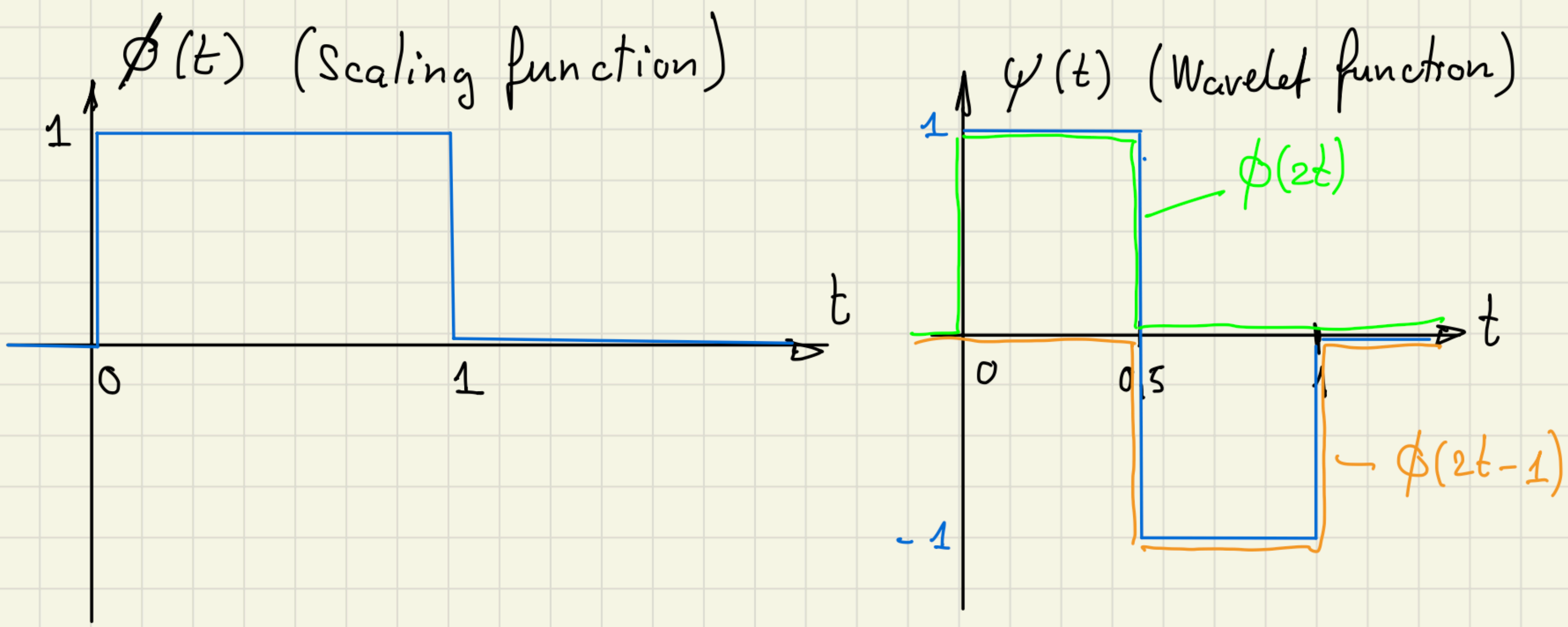
- Inversely we could retrieve the y signal using the synthesis formula: $M^{-1}\alpha = M^H\alpha = y$

Limitations of Fourier Basis

- Fourier Basis is a infinite support orthonormal basis.
  - Adapted for infinite duration signals or periodic signals
  - In case of discontinuity of the signal, Gibs phenomena.
- Due to the infinite support of Fourier transforms basis, each one with a fix frequency we lose the time.
  - after the transformation we lose the time at which this frequency appear into the signal.
  - If we cut the signal in several sequences (periodogram) the product between the sequences duration and the frequency resolution of the spectrum is a constant value.

Compact support orthonormal basis

- The wavelets are a family of orthonormal functions with finite duration (compact support).
- Example of the Haar wavelets:



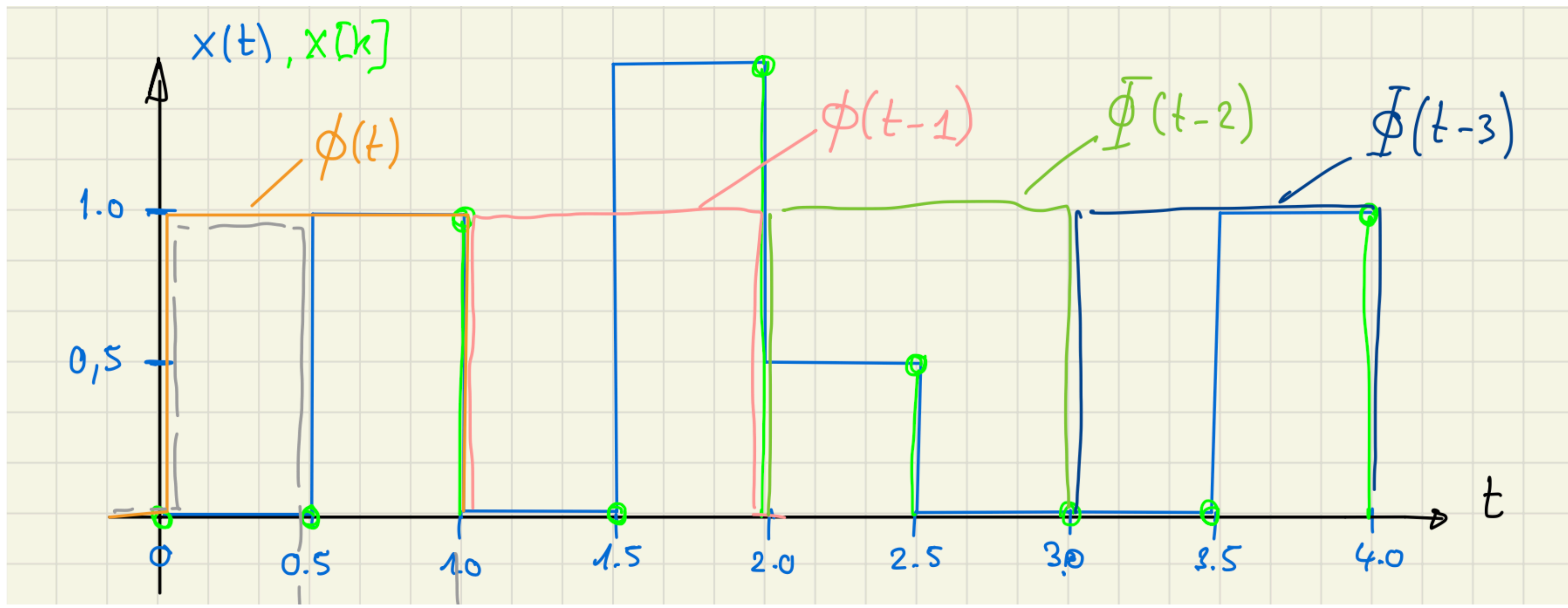
$$\|\phi(t)\|_2 = \langle \phi(t), \phi(t) \rangle = 1 \qquad \|\psi(t)\|_2 = 1$$

$$\langle \phi(t), \psi(t) \rangle = 0 \quad \rightarrow \text{orthogonal}$$

$$\psi(t) = \phi(2t) - \phi(2t-1)$$

Haar wavelet transform

- To illustrate the transform we take a finite sampled signal of N = 8 samples:



Scale coefficients:

$\boxed{a_j[k] = \langle x(t), \phi(t-k) \rangle}$ $\quad k \in 0, 1, 2, \dots$

scaling coeff.

$a_j[0] = \langle x(t), \phi(t) \rangle = 0.50$

$a_j[1] = \langle x(t), \Phi(t-1) \rangle = 0.75$

$a_j[2] = \langle x(t), \Phi(t-2) \rangle = 0.25$

$a_j[3] = \langle x(t), \phi(t-3) \rangle = 0.50$

Detail coefficients:

$\boxed{d_j[k] = \langle x(t), \psi(t-k) \rangle}$ $\quad k \in 0, 1, 2, 3, 4$

$d_j[0] = \langle x(t), \psi(t) \rangle = -0.50$

$d_j[1] = \langle x(t), \psi(t-1) \rangle = -0.75$

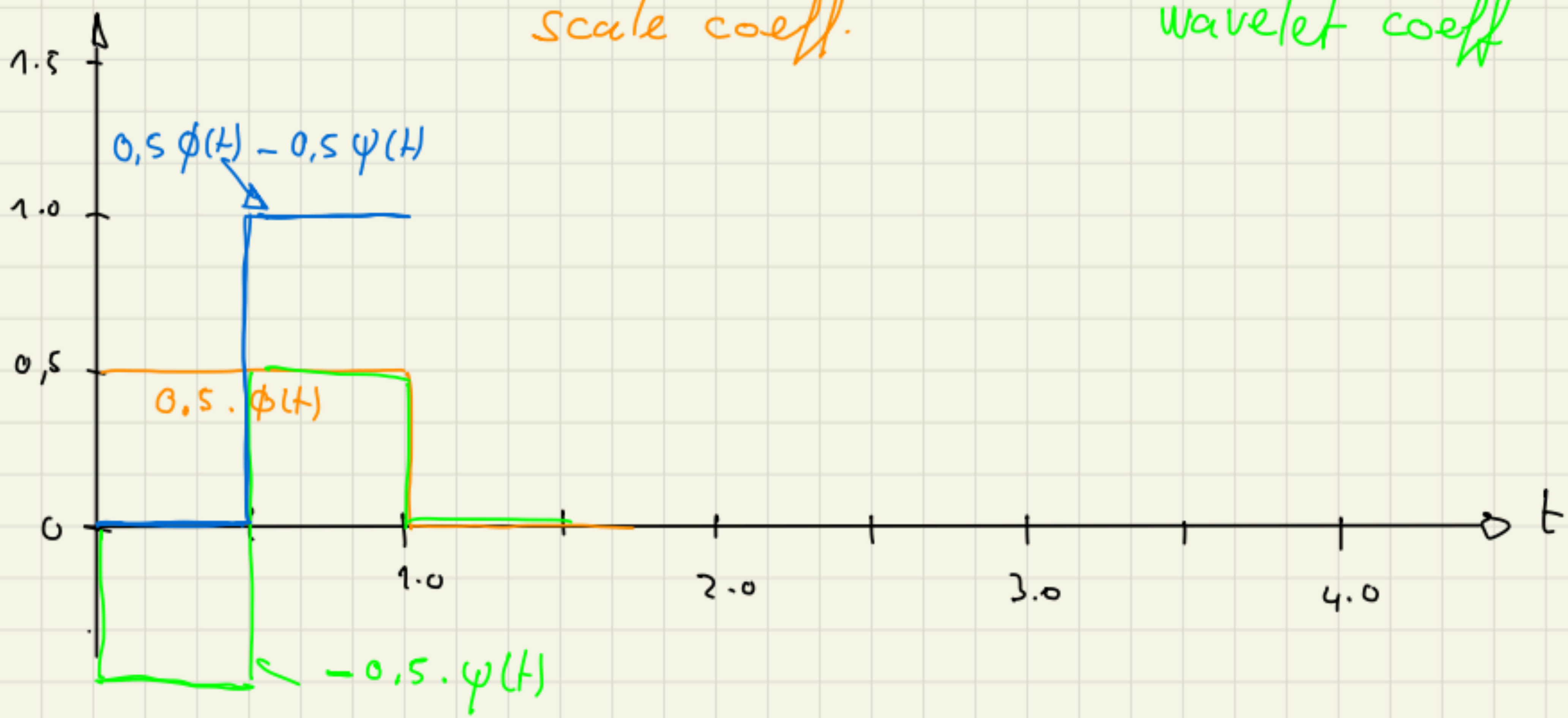
$d_j[2] = \langle x(t), \psi(t-2) \rangle = +0.25$

$d_j[3] = \langle x(t), \psi(t-3) \rangle = -0.50$

Reconstruction

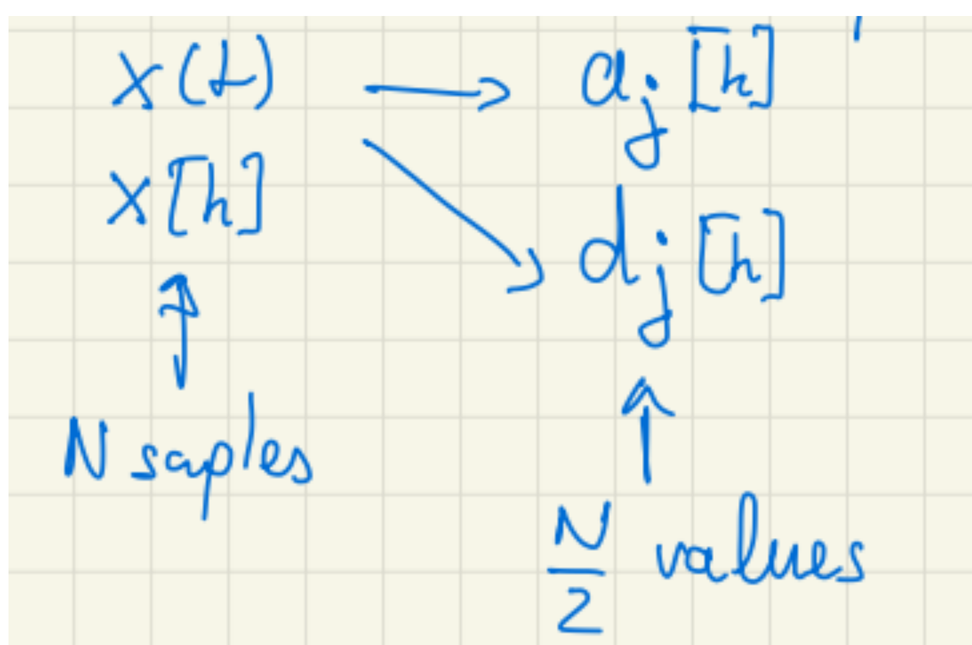
- From the detail coeff. dj[k] and scale coeff. aj[k] we reconstruct the initial signal x(t).

$$x(t) = \sum_k \underbrace{a_j[k]}_{\text{Trend scale coeff.}} \cdot \Phi(t-k) + \sum_k \underbrace{d_j[k]}_{\text{Detail wavelet coeff}} \cdot \psi(t-k)$$



Multiresolution

- In order to represents the details (fastest variations) to the slowest characteristics of the signal we use the multi resolution.
- After the first step of the decomposition from a signal x(t) or a sampled signal x[n] with N samples we obtain set of coefficients aj[k] and dj[k] with N/2 values each.

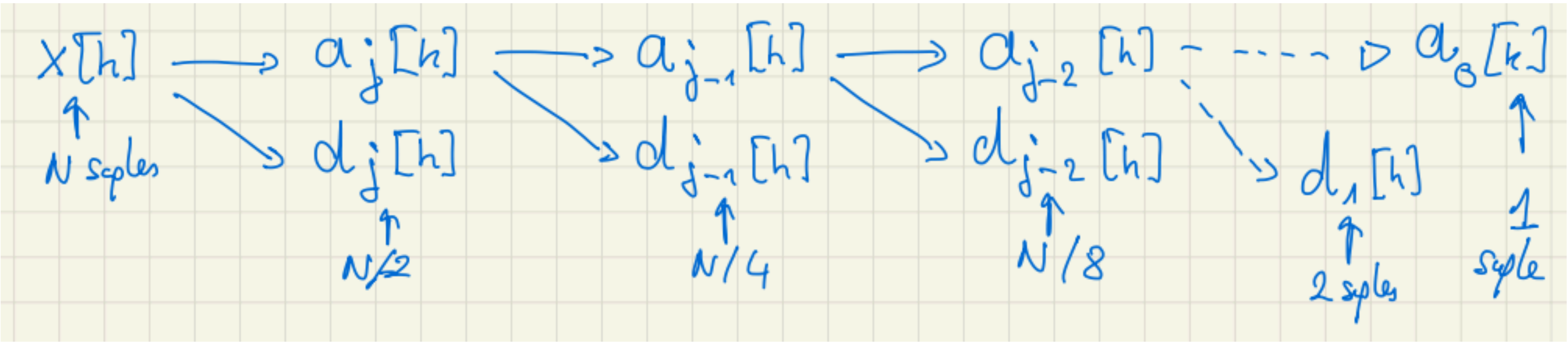


Recursive Multiresolution

- For the next step we consider the coefficient aj[k] as a new signal with N/2 samples.
- We compute a new set of coefficients a(j-1)[k] and d(j-1)[k] decomposing the "signal" aj[k] with the scale function and the wavelet function:

$a_{j-1}[k]$ with the $\Phi[k] = [1, 1]$

$d_{j-1}[k]$ with the $\varphi[k] = [+1, -1]$

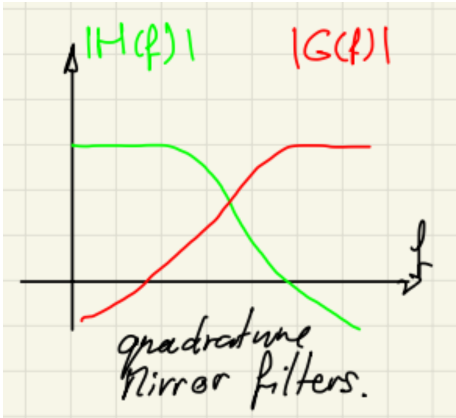


Compute wavelet coeff. without wavelet

- The previous algorithm is in fact a composition of the the discrete convolution and the decimation.
- Two FIR filter are used each one with his impulse response.
  - A low pass filter H with the impulse response h[n] derived from the scale function phi.
  - An high pass filter G with the impulse response g[n] derived from the wavelet function psi.
  - Those 2 filters have particular properties (quadrature filters) ans they are related with the wavelets function as:

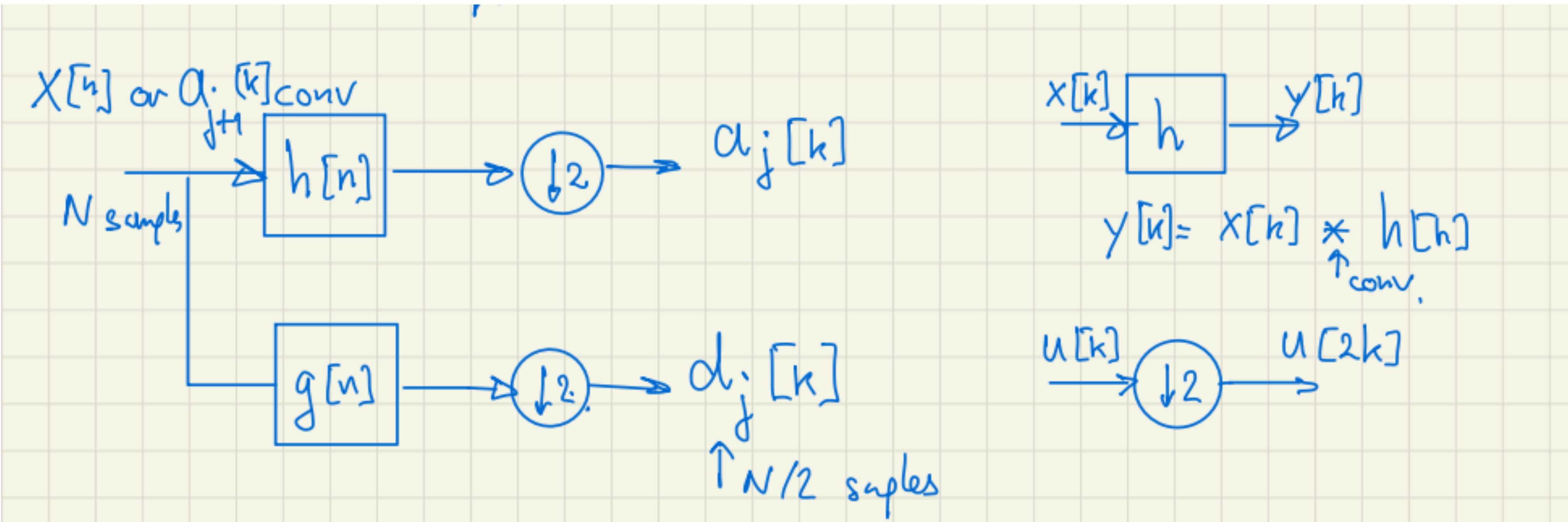
$$g[n] = \frac{1}{\sqrt{2}} \langle \psi(x/2), \phi(x-n) \rangle$$

$$h[n] = \frac{1}{\sqrt{2}} \langle \phi(x/2), \psi(x-n) \rangle$$



Fast Wavelet Transform FWT

- The computation of the Wavelet coefficients aj[k] and dj[k] for every stage J are computed by FWT:



$$\boxed{\begin{aligned} a_j[k] &= (a_{j+1} * h)\downarrow 2 \\ d_j[k] &= (a_{j+1} * g)\downarrow 2 \end{aligned}}$$

a_{j+1} previous with 2^{j+1} samples
or $x[n]$ with N samples

Backward Wavelet Transform

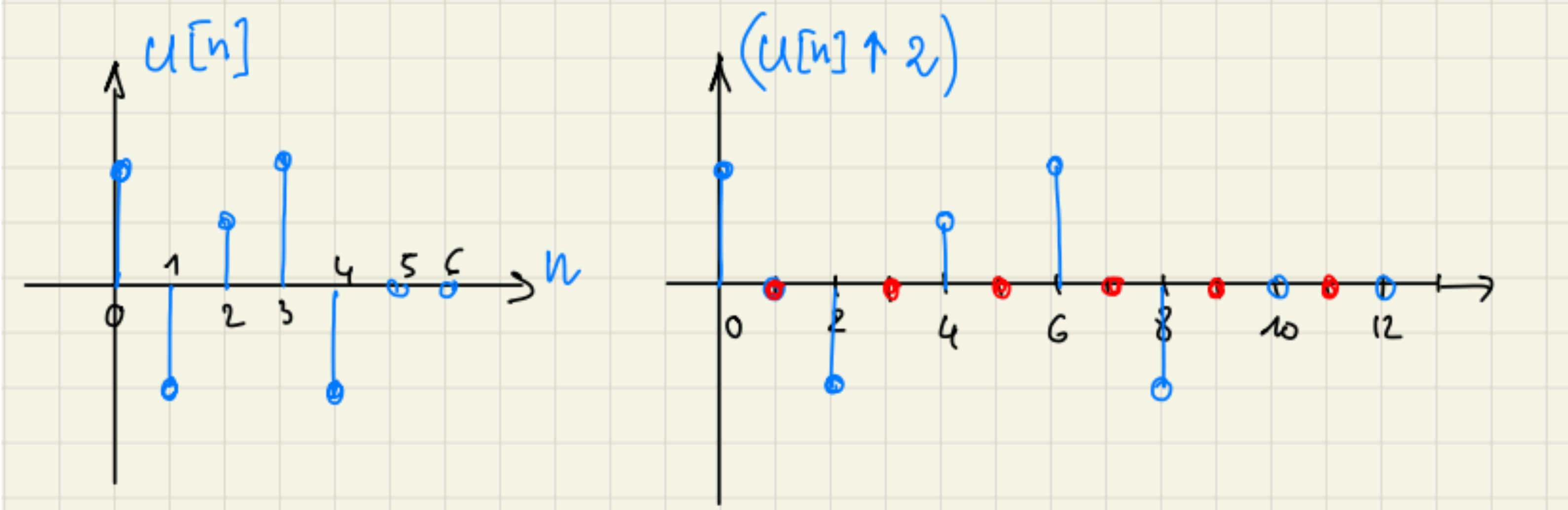
- The backward wavelet transform reconstruct the signal f[n] from the coefficients dj[n] using each scale j from 0 where a0 is a single real constant.
- The recursive algorithm from scale j to scale j+1 uses upsample by 2 and FIR filter with impulse responses $\bar{h}[n] = h[-n]$ and $\bar{g}[n] = g[-n]$.

$$a_{j+1} = (a_j \uparrow 2) * \bar{h}[n] + (d_j \uparrow 2) * \bar{g}[n]$$

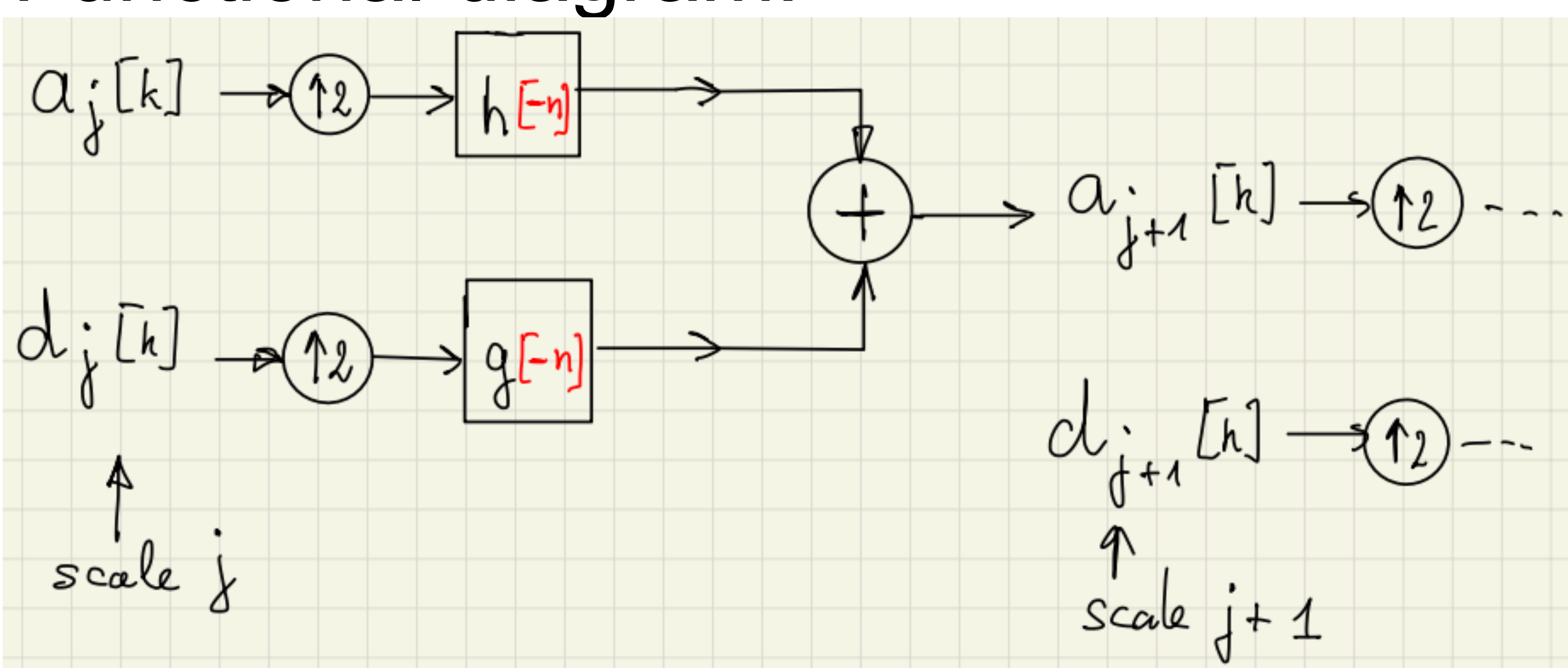
Backward Wavelet Transform

Upsampling: $\left(u[n] \uparrow 2\right)[2n] = u[n]$

$\left(u[n] \uparrow 2\right)[2n+1] = \emptyset$



- Functional diagram:



Other wavelets functions

- Daubechies wavelets is a generalisation of the Haar wavelets, by using longer filters impulse response, producing smother scale function ϕ and wavelets ψ.
- Larger number of coefficients p=2M of the filter , the higher is the number M of vanishing moments.
  - Choosing the best wavelet, and thus choosing M, adapted to given class of signals.
    1. The filter with M=1 vanishing moments correspond to the Haar filter.
    2. The filter with M=2 vanishing moments corresponds to the famous D4 wavelets , which compresses perfectly linear signals
    3. The filter with M=3 vanishing moments compresses perfectly quadratic signals.

Applications

- The first application is the compression of signals or images:
  - After FWT we conserve only the coefficients of each scale j greater than a threshold |dj| > Th anew pu the other to 0.
  - During the reconstruction with backward transform we reconstitue an approximation of the original signal and the error is greater if Th is greater.
  - Another method is to use only a part of the scales from 0 to J_max.
- Today the main use is for the images with 2D FWT, and the JPEG 2000 compression use it.

Applications

- Another application is the de-noising of signals or images :
  - Depending of the nature of the noise we could conserve only a part of the dj coefficients. For exemple to suppress HF noise we could use only dj coefficients with j< J_max.
  - Some wavelets functions are able to suppress constants (offsets) , or polynomial with degree k (moments) , for example linear trends….